

Product description:

- The system shall allow users to paste a suspicious URL and receive a risk assessment.
- It should be very straight forward, the homepage should have a search bar where the user is prompted to insert the link to a suspicious website

Our Minimum viable product (Demo):

- User will paste suspicious URL
- The system scans the URL
- Gives a score (Low/Medium/High) or possibly a number scale (1-10 or 1-100)
- Shows user why it may be suspicious
- Provide tips to users about phishing and the risks

Functional requirements:

- FR-01: The system shall allow users to input a URL for phishing analysis.
- FR-02: The system shall validate whether a submitted URL is in a proper format before processing.
- FR-03: The system shall analyze a submitted URL and return a phishing risk classification.
- FR-04: The system shall classify URLs as Low Risk, Medium Risk, or High Risk.
- FR-05: The system shall provide an explanation of why a URL received its risk classification.
 - EX: Suspicious domain spelling, HTTP instead of HTTPS, Suspicious keywords

User Interface/ Website:

- FR-06: The system shall display scan results on a results page or results section after analysis.
- FR-07: The system shall allow users to submit another URL after receiving results.
- FR-08: The system shall display an error message when invalid input is submitted.
- FR-09: The system shall allow users to report suspected phishing URLs.

Potential Goals:

- FR-10: The system shall allow users to create accounts.
- FR-11: The system shall integrate with a browser extension for real-time phishing detection.
 - EX: Chrome extension, Firefox extension

User Stories:

- US-01: As a user, I want to paste a suspicious URL into a scanner so that I can determine whether the website may be unsafe.
- US-02: As a user, I want the system to validate my URL input so that I know whether I entered a valid website link.
- US-03: As a user, I want to receive a phishing risk level after scanning a URL so that I can decide whether the website is safe to visit.

- US-04: As a user, I want to understand why a URL was flagged as suspicious so that I can learn how to identify phishing attempts.
- US-05: As a user, I want to scan multiple URLs so that I can check more than one suspicious website.
- US-06: As a user, I want to learn common phishing indicators so that I can better recognize malicious websites.
- US-07: As a user, I want phishing prevention recommendations so that I know what actions to take after receiving a risk result.
- US-08: As a user, I want a simple and easy to use interface so that I can quickly scan suspicious URLs.
- US-09: As a user, I want to navigate between website sections so that I can access scanning tools and educational resources/Tips.
- US-10: As a user, I want clear error messages when something goes wrong so that I know how to fix the issue.

Potential User stories:

US-11: As a user, I want to report suspicious URLs so that potentially harmful websites can be documented.

US-12: As a user, I want confirmation after submitting a phishing report so that I know my submission was received.

Future Browser Extension User stories:

US-13: As a user, I want to receive warnings before opening suspicious websites so that I can avoid phishing attacks.

US-15: As a user, I want a browser extension to automatically analyze suspicious links so that I do not have to manually scan them.

SEE QA TEST CASES ON NEXT PAGE:

QA Test Cases:

Test case ID	Feature	Scenario	Test Steps	Expected results
TC-01	URL Input	Submit a valid URL	Enter in a sample URL	The system accepts the URL and begins the analysis
TC-02	URL Validation	Submit empty input	Leave the field empty and click the scan button	User is not allowed to submit an empty URL
TC-03	URL Validation	Submit invalid text	Enter a fake link and click the scan button	System displays invalid URL error
TC-04	URL Validation	Submit URL without HTTPS	Enter a link but with "HTTP:" as opposed to HTTPS	System accepts the URL but flags it as a risk
TC-05	Risk Classification	Scan low risk URL	Submit a trusted URL	System displays low risk
TC-06	Risk Classification	Scan a suspicious URL	Submit a URL with a suspicious keywords	System displays a medium or high risk
TC-07	Results Displays	View scan result	Submit valid URL	System displays risk level
TC-08	Explanation Display	Shows reason for flag	Submit suspicious URL	System explains why URL may be unsafe
TC-09	Error Handling	Extremely long URL	Paste a very long URL and click scan	System handles input without crashing
TC-10	Performance	Scan response time	Submit valid URL	Results appear in a reasonable

				amount of time